

BUSINESS REQUIREMENTS DOCUMENT

Visitor Management System

Analysis & Digital Transformation

A Business Analysis Consulting Engagement

Prepared by

Allulu Alkheraiji

Business Analyst

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0.1	Week 1	Allulu Alkheraiji	Initial draft – problem statement and scope
0.5	Week 2	Allulu Alkheraiji	Added stakeholder analysis, as-is process, pain points
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Approvals

Role	Name / Title	Approval
Project Sponsor	VP, Corporate Operations	Pending
Business Owner	Head of Facilities Management	Pending
Security Lead	Head of Corporate Security	Pending
IT / Solution Owner	Enterprise Applications Manager	Pending

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1. Executive Summary

This Business Requirements Document presents the findings and recommendations of a Business Analysis engagement focused on the organization's visitor management process. The engagement was commissioned in response to recurring complaints regarding long waiting times at reception, inconsistent security checks, and the absence of reliable visitor data for management reporting.

The current visitor management process is predominantly manual. It relies on email exchanges, phone calls, paper logbooks, and disconnected spreadsheets to coordinate visits between visitors, employee hosts, reception staff, and security personnel. This fragmented approach produces approval bottlenecks, poor visibility of who is on site at any given time, inaccurate record keeping, and a visitor experience that does not reflect the organization's brand.

Analysis of the as-is process identified an average visitor processing time of approximately **12–18 minutes** per visitor at peak hours, an approval cycle that frequently exceeds **4 hours**, and a security compliance rate (verified identity and authorized host) estimated at only **60–70%**. These inefficiencies carry measurable cost in lost employee productivity, security exposure, and reputational risk.

This document recommends the design and implementation of a digital Visitor Management System (VMS) providing online pre-registration, an automated approval workflow, QR-code visitor passes, self-service digital check-in and check-out, real-time tracking of on-site visitors, and management dashboards. Benchmarking against comparable enterprise deployments indicates that a digital VMS can reduce visitor processing time by 60–75%, cut approval cycle time by more than 80%, and raise security compliance above 95%.

The expected benefits include a faster and more professional visitor experience, stronger physical security and audit readiness, reduced administrative burden on reception and host employees, and reliable data to support facilities planning and executive reporting. The remainder of this document details the stakeholder landscape, current state, requirements, gap analysis, KPI framework, and an implementation roadmap with associated risks and mitigations.

2. Business Problem Statement

The organization lacks a centralized, digital, and auditable process for managing visitors across its facilities. Visit requests, approvals, and arrivals are coordinated through informal channels — email, phone, instant messaging, and paper logbooks — with no single source of truth.

Symptoms observed

- Visitors experience long and unpredictable wait times at reception, particularly during morning peak hours and large meetings.
- Host employees spend time manually coordinating approvals, notifying security, and escorting visitors, diverting them from core work.
- Security personnel cannot reliably confirm, in real time, who is on site, which host authorized them, or when they are expected to leave.

- Visitor records are incomplete, inconsistent, and difficult to retrieve for audits, incident investigation, or emergency evacuation roll-call.
- Management has no reliable data on visitor volumes, peak periods, or department-level demand to inform staffing and facilities decisions.

Problem definition

Because the visitor management process is manual, fragmented, and undocumented, the organization suffers from approval bottlenecks, weak security compliance, inaccurate records, and an inconsistent visitor experience — resulting in lost productivity, increased security risk, and an inability to report on or improve visitor operations.

3. Project Objectives

The objectives of this initiative are defined to be specific, measurable, and aligned with operational, security, and experience outcomes:

#	Objective	Success Measure
O1	Digitize and centralize the end-to-end visitor management process	Single platform handling 100% of visits across in-scope sites
O2	Reduce visitor processing time at reception	Average check-in time under 2 minutes
O3	Accelerate and standardize the visit approval workflow	Average approval time under 30 minutes
O4	Strengthen physical security and identity verification	Security compliance rate above 95%
O5	Provide real-time visibility of on-site visitors	Live on-site visitor list available to security at all times
O6	Enable data-driven reporting and decision making	Executive, operations, and security dashboards in production
O7	Improve the overall visitor experience	Visitor satisfaction score of 4.3 / 5 or higher

4. Project Scope

4.1 In Scope

- Online visitor pre-registration by host employees and self-registration by visitors via an invitation link.
- Visit scheduling, including single, recurring, and group visits.
- Multi-level approval workflow with automated routing and escalation.
- Automated notifications (email and SMS) to visitors, hosts, and reception.
- QR-code digital visitor passes and badge printing at reception.
- Self-service and reception-assisted digital check-in and check-out.
- Real-time tracking of on-site visitors and watch-list screening.

- Executive, operations, and security dashboards and standard reports.
- Integration with corporate email/calendar and the employee directory for host validation.

4.2 Out of Scope

- Physical access-control hardware procurement (turnstiles, door controllers); the VMS will integrate with existing systems where available, but hardware replacement is excluded.
- Management of contractors and long-term vendors under separate workforce-management processes.
- Employee time-and-attendance tracking.
- Parking management and vehicle registration (candidate for a future phase).
- Replacement of the corporate identity provider or HR system of record.
- Facilities room-booking functionality beyond linking a visit to an existing meeting.

5. Stakeholder Analysis

The following stakeholders were identified through interviews and process observation. Influence reflects the stakeholder's ability to affect the solution's direction and adoption; interest reflects how strongly the outcome affects their daily work.

Stakeholder	Role	Key Responsibilities	Influence	Interest
Visitor	External guest, candidate, client, or partner	Provides accurate details, completes pre-registration, follows site rules	Low	High
Employee Host	Internal sponsor of the visit	Raises requests, selects visit details, receives and escorts the visitor	Medium	High
Receptionist	Front-desk operator	Verifies identity, issues passes, assists check-in, notifies hosts	Medium	High
Security Officer	Physical security and access control	Screens visitors, monitors on-site presence, manages incidents	High	High
Facilities Team	Building and front-of-house operations	Owns reception operations, manages occupancy and equipment	Medium	Medium
Operations Team	Process and service-delivery owner	Owns process performance, SLAs, and continuous improvement	High	High
Executive Management	Project sponsor and decision maker	Approves funding, sets priorities, consumes management reporting	High	Medium

5.1 Stakeholder Engagement Approach

- **Manage closely (High influence / High interest):** Security Officer and Operations Team — involved in design workshops, requirement validation, and user-acceptance testing.
- **Keep satisfied (High influence / Medium interest):** Executive Management — engaged through milestone briefings, business-case validation, and dashboard previews.

- **Keep informed (Medium influence / High interest):** Employee Hosts and Receptionists — engaged through usability testing, training, and feedback loops.
- **Monitor (Low influence / High interest):** Visitors — represented through experience metrics and post-visit feedback rather than direct project participation.

6. Current State Analysis (As-Is)

6.1 Current Process Description

Today a visit begins when a host employee decides to invite an external party. The host typically emails or calls reception to announce the visitor, sometimes copying their manager for approval. Reception records the expected visitor in a shared spreadsheet or a paper logbook. On the day of the visit, the visitor arrives at the front desk, is registered manually, and waits while reception phones the host to confirm and escort them. Security screening, where it occurs, is inconsistent and rarely recorded against the visit.

6.2 Current Workflow

The as-is flow can be summarized in the following sequence:

Visitor Request → Employee Coordination → Email Approval → Security Notification → Visitor Arrival → Manual Registration → Building Access

A detailed swim-lane representation of this workflow is provided in the accompanying As-Is Process Flow deliverable.

6.3 Current Pain Points

- No standard channel for raising a visit request; requests arrive by email, phone, or chat.
- Approvals are informal, untracked, and frequently delayed pending the approver's availability.
- Reception re-keys the same visitor details multiple times across spreadsheets and logbooks.
- Security cannot see a consolidated, real-time list of who is currently on site.
- Visitor records are dispersed and cannot be searched or audited efficiently.
- No data exists to analyze visitor volumes, peak times, or no-show rates.

6.4 Current Operational Challenges

- Peak-hour congestion at reception creates queues and a poor first impression.
- Host productivity is eroded by manual coordination and escorting.
- Inconsistent identity verification exposes the organization to security and compliance risk.
- Manual records undermine emergency evacuation roll-call and incident investigation.
- Lack of metrics prevents evidence-based staffing and facilities planning.

7. Pain Point Analysis

Each pain point is assessed for business impact, severity, and downstream consequence.

Problem	Impact	Severity	Business Consequence
Fragmented request channels	No single intake; requests lost or duplicated	High	Missed visits, rework, poor accountability
Slow, informal approvals	Approval cycle frequently exceeds 4 hours	High	Bottlenecks, delayed or unapproved visits
Manual reception registration	12–18 min processing per visitor at peak	High	Queues, poor experience, host idle time
No real-time visitor visibility	Security blind to who is on site	Critical	Security exposure, failed evacuation roll-call
Inaccurate, dispersed records	Data spread across logbooks and sheets	Medium	Audit failures, no traceability
Inconsistent identity checks	Compliance estimated at 60–70%	Critical	Unauthorized access, regulatory risk
No reporting or analytics	No visibility of volumes or trends	Medium	Reactive staffing, no improvement baseline

8. Root Cause Analysis

8.1 Five Whys Analysis

Problem statement: Visitors wait too long and security cannot confirm who is on site.

1. Why do visitors wait too long? Because reception must manually register each visitor and phone the host on arrival.
2. Why is registration manual and the host phoned on arrival? Because there is no pre-registration or automated notification before the visit.
3. Why is there no pre-registration or automated notification? Because there is no shared system to capture visit details in advance.
4. Why is there no shared system? Because the process grew informally and was never digitized or assigned an owner.
5. Why was it never digitized or owned? Because visitor management was treated as a low-priority administrative task rather than a measurable business process.

Root cause: *The absence of an owned, digital, end-to-end visitor management process — caused by treating visitor handling as informal administration rather than a managed operational process — is the root of the wait times, weak security compliance, and lack of data.*

8.2 Fishbone (Ishikawa) Analysis Summary

Contributing causes were grouped into six categories. The summary below captures the principal drivers under each.

Category	Contributing Causes
People	Hosts coordinate manually; reception re-keys data; security checks depend on individual diligence; no clear process owner
Process	No standard intake; informal untracked approvals; no pre-registration; no defined SLAs
Technology	No VMS; reliance on email, phone, spreadsheets, and paper logbooks; no integration with directory or calendar
Data	Records dispersed and inconsistent; no unique visit identifier; no analytics or reporting
Environment	Single congested reception point; peak-hour surges; multiple entrances with uneven coverage
Measurement	No KPIs; processing, approval, and compliance times never measured or reported

9. Gap Analysis

The table contrasts the current state with the target future state and identifies the gap the VMS must close.

Dimension	Current State (As-Is)	Future State (To-Be)	Gap Severity
Visit request	Email, phone, chat — no standard	Single online form / invitation link	High
Approval	Informal, > 4 hours	Automated workflow, < 30 min	High
Registration	Manual, 12–18 min	Digital self check-in, < 2 min	High
Visitor pass	Handwritten or none	QR-code digital pass + badge	Medium
On-site visibility	None / paper logbook	Real-time on-site dashboard	Critical
Identity / security	Inconsistent, 60–70%	Verified + watch-list, > 95%	Critical
Records	Dispersed, non-auditable	Centralized, searchable, auditable	High
Reporting	None	Executive / ops / security dashboards	Medium
Notifications	Manual phone calls	Automated email + SMS	Medium

10. Future State Design (To-Be)

The proposed Visitor Management System digitizes the full visitor lifecycle from invitation to departure. Its core capabilities are described below; the end-to-end flow is illustrated in the accompanying To-Be Process Flow deliverable.

Online Visitor Registration

Hosts pre-register visitors in seconds; visitors complete their own details, consent, and document upload through a secure invitation link before arrival.

Visit Scheduling

Single, recurring, and group visits can be scheduled and linked to a calendar meeting, with capacity awareness for large events.

Approval Workflow

Configurable multi-level approval routes requests to the right approver automatically, with reminders and escalation to prevent bottlenecks.

Automated Notifications

Visitors, hosts, and reception receive timely email and SMS updates: invitation, confirmation, reminders, arrival alerts, and check-out confirmation.

QR-Code Visitor Pass

On approval, each visitor receives a unique QR-code pass for fast, contactless check-in and optional integration with access-control gates.

Digital Check-In / Check-Out

Visitors self-check-in via kiosk or QR scan; reception can assist where needed. Check-out is captured automatically or on pass scan at exit.

Real-Time Visitor Tracking

Security sees a live list of everyone on site, their host, location, and expected departure, with watch-list screening and over-stay alerts.

Dashboard & Reporting

Role-based dashboards deliver executive trends, operational throughput, and security status, supported by exportable standard reports.

11. Business Requirements

Business requirements express what the organization needs the solution to achieve, independent of implementation.

ID	Business Requirement	Priority
BR-001	Provide a single, centralized platform to manage all visitors across in-scope sites.	High
BR-002	Enable host employees to pre-register visitors in advance of arrival.	High
BR-003	Enable visitors to self-complete their details before arriving on site.	High
BR-004	Provide a standardized, auditable visit-approval workflow.	High
BR-005	Reduce average visitor processing time at reception to under two minutes.	High
BR-006	Reduce average visit-approval time to under thirty minutes.	High
BR-007	Verify visitor identity and screen against a watch list before access.	High
BR-008	Provide security with real-time visibility of all visitors on site.	High

ID	Business Requirement	Priority
BR-009	Maintain a complete, searchable, auditable record of every visit.	High
BR-010	Notify visitors, hosts, and reception automatically at each visit stage.	Medium
BR-011	Issue a secure visitor pass enabling fast, controlled building access.	Medium
BR-012	Support emergency evacuation roll-call from live on-site data.	High
BR-013	Provide management dashboards and reports on visitor operations.	Medium
BR-014	Integrate with the employee directory and corporate calendar/email.	Medium
BR-015	Capture visitor consent and handle personal data in line with privacy regulation.	High
BR-016	Support multiple sites, entrances, and visit types from one configuration.	Medium
BR-017	Measure visitor experience through post-visit feedback.	Low

12. Functional Requirements

Functional requirements describe specific system behaviours that satisfy the business requirements.

ID	Functional Requirement	Traces To
FR-001	The system shall allow a host to create a visit by entering visitor name, company, email, mobile, date, time, and purpose.	BR-002
FR-002	The system shall send an invitation link allowing visitors to complete their own registration and upload an ID document.	BR-003
FR-003	The system shall support single, recurring, and group visit creation.	BR-002
FR-004	The system shall route each visit through a configurable approval workflow based on visit type, location, or visitor risk.	BR-004
FR-005	The system shall send reminders and escalate approvals not actioned within a defined time.	BR-006
FR-006	The system shall validate the host against the employee directory before confirming a visit.	BR-014
FR-007	The system shall screen each visitor against a configurable watch list and flag matches.	BR-007
FR-008	The system shall generate a unique QR-code pass upon approval and deliver it to the visitor.	BR-011
FR-009	The system shall allow self check-in via kiosk or QR scan and assisted check-in at reception.	BR-005
FR-010	The system shall record check-out automatically on exit scan or after a configurable time.	BR-009
FR-011	The system shall display a live list of on-site visitors with host, location, and expected departure.	BR-008
FR-012	The system shall raise an alert when a visitor over-stays an expected departure time.	BR-008
FR-013	The system shall send automated email and SMS notifications at invitation, confirmation, arrival, and check-out.	BR-010
FR-014	The system shall produce an emergency on-site roster on demand.	BR-012

ID	Functional Requirement	Traces To
FR-015	The system shall provide role-based dashboards for executives, operations, and security.	BR-013
FR-016	The system shall allow authorized users to search and export visit records by date, host, or visitor.	BR-009
FR-017	The system shall capture and store visitor consent with each registration.	BR-015
FR-018	The system shall send a post-visit feedback request and record the response.	BR-017

13. Non-Functional Requirements

Category	ID	Requirement
Security	NFR-01	All data encrypted in transit (TLS 1.2+) and at rest; role-based access control; full audit logging of every action.
Security	NFR-02	Personal data retained only for a configurable period and purged automatically in line with privacy regulation.
Performance	NFR-03	Check-in transaction completes within 3 seconds; dashboards refresh within 5 seconds for up to 1,000 daily visits.
Availability	NFR-04	Solution available 99.9% during business hours with a defined disaster-recovery and backup plan.
Scalability	NFR-05	Architecture supports growth to multiple sites and 10x current visit volume without redesign.
Usability	NFR-06	Visitor self check-in completed by a first-time user in under 60 seconds without assistance; interface is bilingual (Arabic/English) and accessible.
Compatibility	NFR-07	Web interface works on current versions of major browsers and on tablet kiosks.
Maintainability	NFR-08	Workflows, notifications, and watch lists are configurable by an administrator without code changes.

14. Business Rules

ID	Business Rule
BRULE-01	A visitor may not enter the building without an approved and active visit record.
BRULE-02	Every visit must have a valid internal employee host.
BRULE-03	A visit request must be approved before a QR-code pass is issued.
BRULE-04	Visitor identity must be verified at first check-in against an official ID.
BRULE-05	Any watch-list match must block automatic approval and require security review.
BRULE-06	An approval not actioned within the defined SLA escalates to the next approver.
BRULE-07	A visitor pass is valid only for the scheduled visit date and times.
BRULE-08	A visitor not checked out by end of day is auto-checked-out and flagged for review.
BRULE-09	Visitor personal data must be purged after the defined retention period.

ID	Business Rule
BRULE-10	Only authorized security and operations roles may view the full on-site visitor list.
BRULE-11	Group visits require a single responsible host accountable for all attendees.

15. KPI Framework

The following KPIs will measure the performance of the visitor management process before and after implementation.

KPI	Definition	Formula	Target	Purpose
Total Visitors	Count of visits in a period	Σ visits in period	Track	Demand & capacity planning
Visitor Approval Time	Time from request to approval	Approval timestamp – request timestamp	< 30 min	Identify approval bottlenecks
Visitor Processing Time	Time from arrival to host meeting	Host-met time – arrival time	< 5 min	Measure reception efficiency
Check-In Time	Duration of the check-in step	Check-in complete – check-in start	< 2 min	Optimize the front-desk experience
No-Show Rate	Share of approved visits that never arrive	$(\text{No-shows} \div \text{approved visits}) \times 100$	< 10%	Reduce wasted preparation
Visitor Satisfaction	Average post-visit rating	Σ ratings $\div$ responses	$\geq 4.3 / 5$	Track experience quality
Security Compliance Rate	Visits with verified ID and authorized host	$(\text{Compliant visits} \div \text{total}) \times 100$	> 95%	Reduce security risk

16. Dashboard Requirements

Three role-based Power BI dashboards are proposed. A detailed visual mockup is provided in the Power BI Dashboard deliverable.

16.1 Executive Dashboard

- **KPIs:** Total Visitors, visitor trend vs. prior period, peak visit hours, visits by department.
- **Charts:** KPI cards, line chart (trend), heat map (peak hours), bar chart (department).
- **Filters:** Date range, site, department, visit type.
- **Layout:** KPI cards across the top, trend line and peak-hour heat map in the middle, department bar chart at the foot.

16.2 Operations Dashboard

- **KPIs:** Pending approvals, upcoming visits today, average visitor processing time, no-show rate.
- **Charts:** Pending-approvals card with drill-through, upcoming-visits table, gauge for processing time.
- **Filters:** Date, site, host, status.
- **Layout:** Action cards on the left, today's visit schedule table on the right, processing-time gauge below.

16.3 Security Dashboard

- **KPIs:** Current visitors on site, expired/over-stay passes, security alerts, watch-list flags.
- **Charts:** Live on-site count card, on-site visitor table, alert list, compliance gauge.
- **Filters:** Site, entrance, time window, alert type.
- **Layout:** Live on-site count and compliance gauge at the top, on-site visitor table centre, alerts panel right.

17. Process Improvement Recommendations

1. Implement a single digital intake channel for all visit requests, eliminating email and phone coordination.
2. Introduce mandatory pre-registration so visitor data is captured before arrival, not at the desk.
3. Automate the approval workflow with defined SLAs, reminders, and escalation to remove bottlenecks.
4. Deploy self-service check-in via kiosk and QR code to cut reception processing time.
5. Integrate host validation with the employee directory to guarantee a valid sponsor for every visit.
6. Enable real-time on-site tracking and over-stay alerts to strengthen security and evacuation readiness.
7. Standardize identity verification and watch-list screening at first check-in.
8. Establish the KPI framework and dashboards as the baseline for continuous improvement.
9. Assign a clear process owner (Operations) accountable for visitor-management performance.
10. Automate post-visit feedback to monitor and improve the visitor experience.
11. Roll out in phases — pilot at one site, refine, then scale — to manage change and risk.
12. Provide role-based training and quick-reference guides to drive adoption among hosts and reception.

18. Expected Benefits

Benefits are quantified by comparing the measured current state with benchmarked target performance.

Benefit	Current	Target	Improvement
Visitor processing time	12–18 min	< 2 min	~85% reduction
Visit approval time	> 4 hours	< 30 min	> 80% reduction
Security compliance rate	60–70%	> 95%	+25–35 points
Visitor satisfaction	Not measured	≥ 4.3 / 5	New baseline & uplift
Reception admin effort	High, manual	Largely automated	~60% effort saved
On-site visibility	None	Real-time	Full traceability
Management reporting	None	Live dashboards	Data-driven decisions

19. Risks & Mitigation Plan

Key delivery and adoption risks are assessed by likelihood and impact, with mitigation owners assigned.

ID	Risk	Likelihood	Impact	Mitigation
R1	Low user adoption by hosts and reception	Medium	High	Early involvement, training, simple UX, executive sponsorship
R2	Integration issues with directory / calendar / access control	Medium	High	Early technical discovery, phased integration, fallback manual mode
R3	Data privacy / consent non-compliance	Low	High	Privacy-by-design, consent capture, retention rules, legal review
R4	Scope creep (parking, contractors)	Medium	Medium	Clear scope baseline, change control, future-phase backlog
R5	Peak-hour performance shortfall	Low	Medium	Load testing, scalable architecture, kiosk capacity planning
R6	Resistance from security to process change	Medium	Medium	Co-design with security, pilot validation, demonstrate visibility gains
R7	Data migration / historical record gaps	Low	Low	Start clean, archive legacy logs, define go-live cut-over

20. Conclusion

The current visitor management process is manual, fragmented, and unmeasured, producing slow approvals, long reception queues, weak security compliance, and an absence of reliable data. These issues carry tangible cost in productivity, security exposure, and visitor experience.

The recommended digital Visitor Management System directly addresses each root cause identified in this analysis. By centralizing intake, automating approvals, digitizing check-in, providing real-time visibility, and delivering management dashboards, the organization can reduce visitor processing time by roughly 85%, cut approval time by over 80%, and raise security compliance above 95% — while establishing, for the first time, a measurable baseline for continuous improvement.

A phased implementation — beginning with a single-site pilot — will manage change and de-risk delivery. With executive sponsorship, cross-functional engagement, and the KPI framework defined in this document, the initiative is positioned to deliver a faster, safer, and more professional visitor experience aligned with the organization’s operational and security objectives.

Appendix A: Requirements Traceability Matrix

The matrix links objectives to business and functional requirements, demonstrating full coverage.

Objective	Business Requirements	Functional Requirements
O1 Digitize & centralize	BR-001, BR-016	FR-001, FR-003, FR-016
O2 Reduce processing time	BR-005, BR-011	FR-008, FR-009, FR-010
O3 Accelerate approvals	BR-004, BR-006	FR-004, FR-005, FR-006
O4 Strengthen security	BR-007, BR-015	FR-007, FR-017
O5 Real-time visibility	BR-008, BR-012	FR-011, FR-012, FR-014
O6 Reporting	BR-013	FR-015, FR-016
O7 Visitor experience	BR-010, BR-017	FR-002, FR-013, FR-018